DEPARTMENT OF THE NAVY
NAVAL SURFACE WARFARE CENTER
PHILADELPHIA DIVISION
5001 SOUTH BROAD STREET
PHILADELPHIA PA 19112-1403

IN REPLY REFER TO:

J&A No. 0241-24-129-00
Code 0241
PID No. N64498-23-RFPREQ-PD-45-0009

BRAND-NAME JUSTIFICATION

1. Contracting Activity

The Naval Sea Systems Command, Naval Surface Warfare Center, Philadelphia Division (NSWCPD), Code 0241.

2. Description of Action Being Approved

Award of a Firm-Fixed-Price (FFP), single award Indefinite-Delivery Indefinite-Quantity (IDIQ) contract for fabrication of Tow Cable Assemblies only sourced from brand-name components from Rochester Wire & Cable, LLC, Seacon Corporation, and PMI Industries, Inc.

3. Description of the Supplies or Services

The contractor shall fabricate, assemble, test, and deliver roughly forty (40) cable assemblies per year for the OHIO and COLUMBIA Class submarines during the proposed contract's three (3) year period. Each cable assembly shall consist of brand-name components from Rochester Wire & Cable, LLC; Seacon Corporation; and PMI Industries Inc. The table below is a breakdown of the value for the brand-name components:

Part Number	Item Description	Manufacturer	Estimated Unit Cost	Estimated Total Quantity	Estimated Total Cost
A309011 (OHIO)	Subsea electro-mechanical communications cable.	Rochester Wire & Cable, LLC	[redacted]	42,030 ft. of cable	[redacted]
A304177 (COLUMBIA)	Subsea electro-mechanical communications cable.	Rochester Wire & Cable, LLC	[redacted]	41,880 ft. of cable	[redacted]
MSSK12CCP/MSSL12CCP (OHIO)	Dry-mate electrical cable connectors.	Seacon Corporation	[redacted]	31 units	[redacted]
MSSK12CCP/MSSL12CCP (COLUMBIA)	Dry-mate electrical cable connectors.	Seacon Corporation	[redacted]	34 units	[redacted]
9210041 (OHIO)	Dam/Blok technology – water isolating, depth pressure splice assembly.	PMI Industries, Inc.	[redacted]	31 units	[redacted]
9210041 (COLUMBIA)	Dam/Blok technology – water isolating, depth	PMI Industries Inc.	[redacted]	34 units	[redacted]

	pressure splice assembly.				
				Total:	[redacted]

These items are non-commercial parts. The estimated value of the brand-name components is [redacted]. There are no option quantities or periods. This contract will be funded at the delivery order level. The Government's minimum needs have been verified by the certifying technical and requirements personnel.

Additionally, the table below shows the estimated breakdown of funding, by appropriation type, per year, for the duration of this contract, which reconciles with the Independent Government Cost Estimate (IGCE):

Funding Type	FY	Total
OPN (OHIO)	FY24	[redacted]
SCN (COLUMBIA)	FY24	[redacted]
SCN (COLUMBIA)	FY25	[redacted]
OPN (OHIO)	FY25	[redacted]
OPN (OHIO/COLUMBIA)	FY26	[redacted]
	TOTAL:	[redacted]

*For clarification, the $2M of material is only for the brand-name portion of the tow cables. The additional $2M, which is the prime's work assembling the tow cables, is not sole-source or brand-name.

The Government's minimum needs have been verified by the certifying technical and requirements personnel.

4. <u>Brand-Name Supporting Rationale</u>

FAR 6.302-1, pursuant to 10 U.S.C. 3204(a)(1), Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements.

The use of the referenced brand-name sources for this procurement is necessary to ensure support of the fielding of improved cable assemblies to reduce failure rates of a deployed critical system on OHIO and COLUMBIA Class submarines.

Rochester is the sole vendor that is qualified to manufacture the subsea electro-mechanical communications cable in accordance with the required product specifications (PS8628115 & PS8628116). The Tow Cable Improvements Critical Design Review (CDR) evaluation process documented Rochester's capabilities, qualifications, and quality assurance processes ensuring that the cables produced in their facilities meet all required specifications and requirements.

Rochester's proprietary cable design was achieved through several design iterations, requiring almost four years for research, development, prototyping, and testing. Their manufacturing processes are unique and proprietary. They manufacture raw cable that exclusively meet's the Navy's stringent Submarine cable electronic and mechanical performance specifications. Their internal conductor positioning processes, cable winding and cable casing application are all unique. The cables they produce meet exacting physical dimensional requirements needed for use on existing SSBN Submarine Class Buoy Systems. If other vendors were utilized to manufacture this cable, they would be required to produce a new design including all R&D and testing to down select the appropriate cable designs for this application, which would ultimately be required to be approved by the Government. Utilizing a vendor other than Rochester for the raw tow cable would result in unacceptable schedule and cost impacts and compromise critical ships systems that require the updated cable assemblies.

Seacon owns the proprietary design rights for the electrical terminations/end connectors used in the tow cable assembly and is therefore the sole vendor capable of providing this product in accordance with the specification documents (PS8628115 & PS8628116). Seacon connectors exclusively mate to existing submarine buoy system electronic equipment. The connector design includes proprietary manufacturing molds and manufacturing process that cannot be duplicated without major development costs, production time and test certifications. This proprietary design was evaluated during the test and evaluation of the re-designed tow cable assembly and determined to meet all required product and performance specifications. Seacon's proprietary design was developed specifically for interoperability with Rochester's raw tow cable.

PMI Industries is the only vendor that can provide the bespoke Dam/Blok device. Dam/Blok is a patented/proprietary product developed by PMI. The device integrates into the polyurethane molded sections defined in NAVSEA drawing 8603582 Rev A and drawing PS8628116 Rev (-). The Dam/Blok design and procedure is the only product that meets the military's electronic cable specification for use on submarines. It provides protection to the cables and downstream equipment they supply should cable water intrusion occur, which is a common occurrence, based on the extreme operating environments the cables are subject to. Market research has shown that no other vendor can produce a substitute replacement for this product. Utilizing a vendor other than PMI to provide a replacement for the Dam/Blok device would result in unacceptable schedule and cost impacts and compromise critical ships systems that require the updated cable assemblies.

There would be significant cost and schedule impacts to the Navy should the procurements not require these brand-name components. The current tow cable designs with selected components were the result of a lengthy, iterative design process, comprising of roughly six (6) years of research, development, and testing of the manufacturing processes/procedures required to be approved by the Government. No other vendors can provide the components containing the essential characteristics meeting the Government's requirement as detailed above. Because the tow cable and its components were designed, tested, and evaluated as a complete assembly; any modification to any internal component would result in the re-design, test, and evaluation of the new assembly. If new vendors were to be selected to develop replacement components, it would result in a duplication of cost upwards of $5.6M that would not be recouped through competition and result in unacceptable delays of up to four (4) years putting at risk the delivery of the critical components prior to the ship's planned installation window.

Furthermore, market research was conducted through a survey and analysis of alternative sources available in the industry under Request for Information (RFI) 21-DR-028 in FY21 and again under Notice ID: 0241 in March 2024. The thorough analysis conducted reviewed the available brands to see if there were any possible alternatives. Overall, none of the responses demonstrated the technical capabilities required to meet the established design and performance specifications.

Based on market research, raw cable material from Rochester Wire & Cable, LLC, Dry-Mate Electrical cable connectors from Seacon Corporation, and Dam/Blok™ and technology Splice assembly from PMI Industries, Inc., they are the only sources capable of meeting the minimum requirements.

For these reasons, it is imperative that the Rochester Wire & Cable, LLC, Seacon Corporation, and PMI Industries, Inc. brand-name equipment is specified in the requirements Statement of Work (SOW) as the required components.

5. Determination of Fair and Reasonable Costs

The Contracting Officer has determined the anticipated cost to the Government for the supplies covered by this Justification will be fair and reasonable.

6. Any Other Facts Supporting the Justification

This Justification will be posted along with the solicitation to the NASA SEWP website. Any responses received will be evaluated.

7. Actions to Remove Barriers to Competition

NSWCPD will continue to monitor the marketplace and identify potential sources of the supplies identified in this Justification. Any sources discovered that can potentially satisfy this specific procurement will be considered for any future requirements.